Building Agent Hub with AI

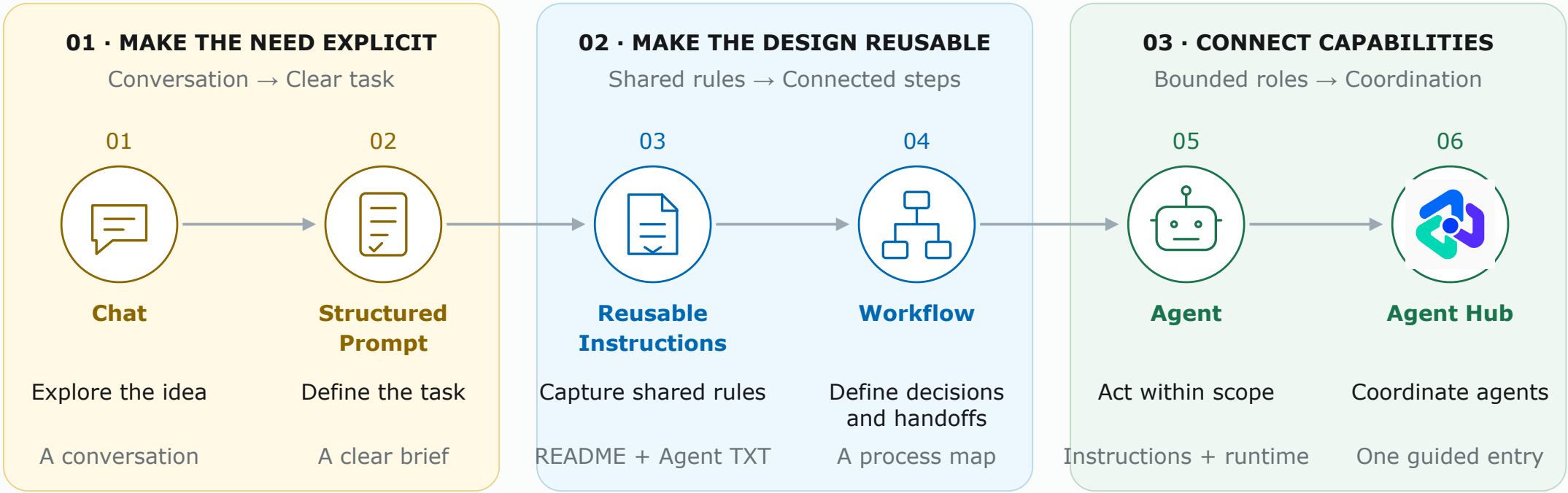
From idea to Files
Communicating with AI, Defining standards, and Iterating

SPEAKER: **XINGYU YOU** DATE: **2026-10-20**

01 · THE BIG PICTURE

From Chat to Agent Hub: the whole journey at a glance

Follow one project from an initial conversation to reusable rules and coordinated agent roles.

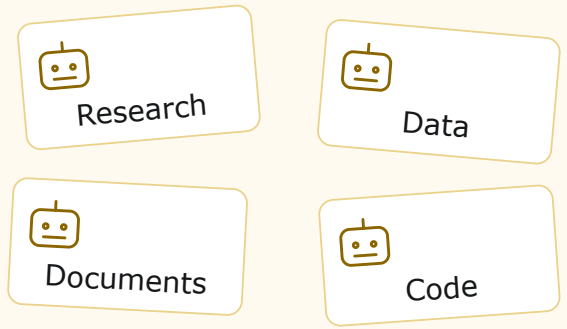


A prompt becomes part of an agent design. Runtime, tools, state handling, and verification make execution possible.

Conceptual roadmap, not a strict creation timeline. Project definitions do not by themselves prove deployed capability.

Why Agent Hub? From too many options to coordinated work

01 · Too many agents



"Which one fits my task?"

The user knows the goal, not the agent.

02 · Recommend and explain

1 **Candidate A** — Strong fit
Capability · input fit · permissions

2 **Candidate B** — Partial fit
Capability · input fit · permissions

3 **Candidate C** — Limited fit
Capability · input fit · permissions

Explain the ranking and limits.

Illustrative shortlist, not actual scores.

03 · Help complete the task

Select agents + plan

↓

User confirms scope

↓

Execute within permission

↓

Return reviewable results

Hub coordinates the handoffs.

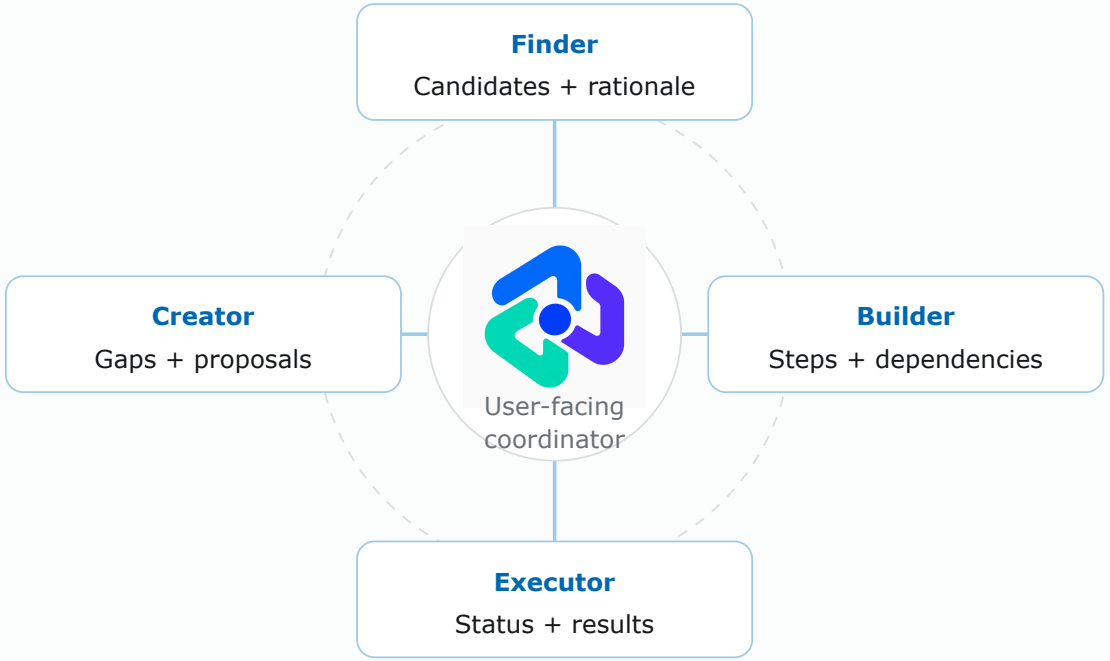
Describe the goal → understand the recommendation → confirm the work. Agent Hub connects the three.

Project design rationale, not completed user research. Candidates and fit levels illustrate the intended experience. Runtime availability, tools, and authorization still require verification.

02 · ROLE DESIGN

How we shaped the main-agent framework with AI

01 **Share the idea** — 02 **Break down the work** — 03 **Assign roles** — 04 **Define boundaries**



ILLUSTRATIVE CONVERSATION · NOT ORIGINAL CHAT

USER · Share the idea (U)

I want a single entry point that recommends agents, explains fit, and helps complete tasks. Help me structure the system first.

AI · Break down the work (AI)

We need request understanding, capability matching, workflow planning, task execution, and capability-gap handling.

USER · Ask for the framework (U)

Which work belongs to the main agent? Which needs a functional agent? Keep planning separate from execution and make boundaries explicit.

AI · Propose the role split (AI)

Hub owns interaction and routing. Finder matches, Builder plans, Executor runs confirmed work, and Creator assesses gaps and reuse.

The framework answers who owns each responsibility. The next slide connects those roles into an initial workflow. Dialogue is reconstructed, not an original transcript.

From roles to an initial workflow



RECONSTRUCTED EXAMPLE · NOT ORIGINAL CHAT

USER · Connect the roles

Using the agreed roles, outline the main path from a user request to a reviewable result.

AI · Propose a main path

Understand the request → find or validate agents → plan if needed → confirm → execute → review.

USER · Challenge the first draft

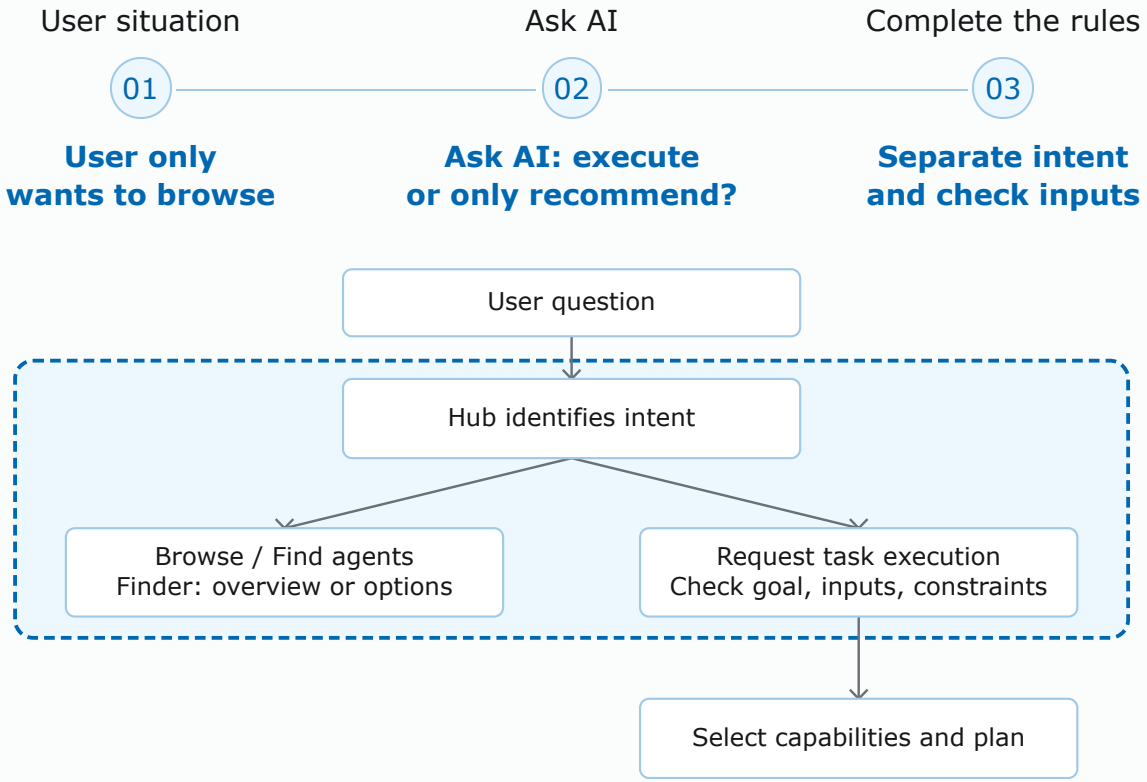
Now examine it from the user's perspective. What different questions or missing inputs could change this path?

DESIGN OUTCOME

The main path is a starting point. Next, user questions reveal the missing branches.

03 · WORKFLOW

User perspective 1: “I am only browsing.”



USER-LED FOLLOW-UP · ILLUSTRATIVE DIALOGUE

USER · Design question (U)

If a user only asks what agents exist, should Hub execute anything?

AI · Proposed rule (AI)

No. Separate browsing from execution. Finder returns an overview or recommendations.

USER · Follow-up (U)

What if the goal is clear, but the user has not supplied the materials?

IMPLEMENTED IN THE DESIGN

Discuss suitable capabilities if useful, but wait for required inputs before execution. Ask focused questions if the goal itself is unclear.

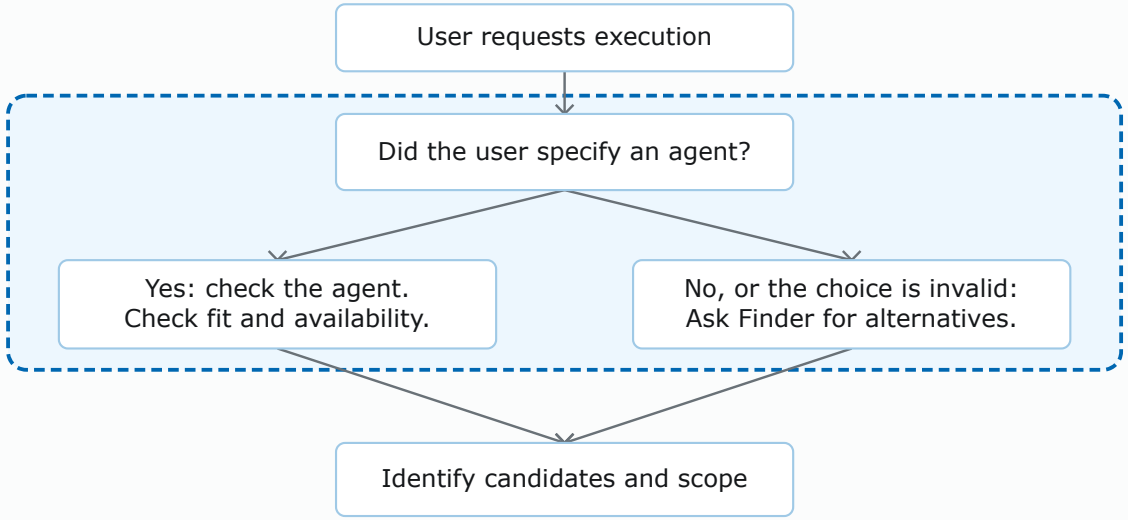
User perspective 2: “I already chose an agent.”

User situation → Ask AI → Complete the rules

01 **User names an agent**

02 **Ask AI: validate or find alternatives?**

03 **Check suitability before proceeding**



USER-LED FOLLOW-UP · ILLUSTRATIVE DIALOGUE

USER · Start with the need

Users may name an agent. Should Hub still ask Finder to recommend one?

AI · Proposed design

Validate the specified agent first. If it fits and can run, skip a fresh recommendation.

USER · Refine the boundary

What if it is invalid? Does choosing an agent authorize execution?

IMPLEMENTED IN THE DESIGN

Invalid choice → explain and ask Finder. Selection alone does not authorize execution. Scope confirmation remains required.

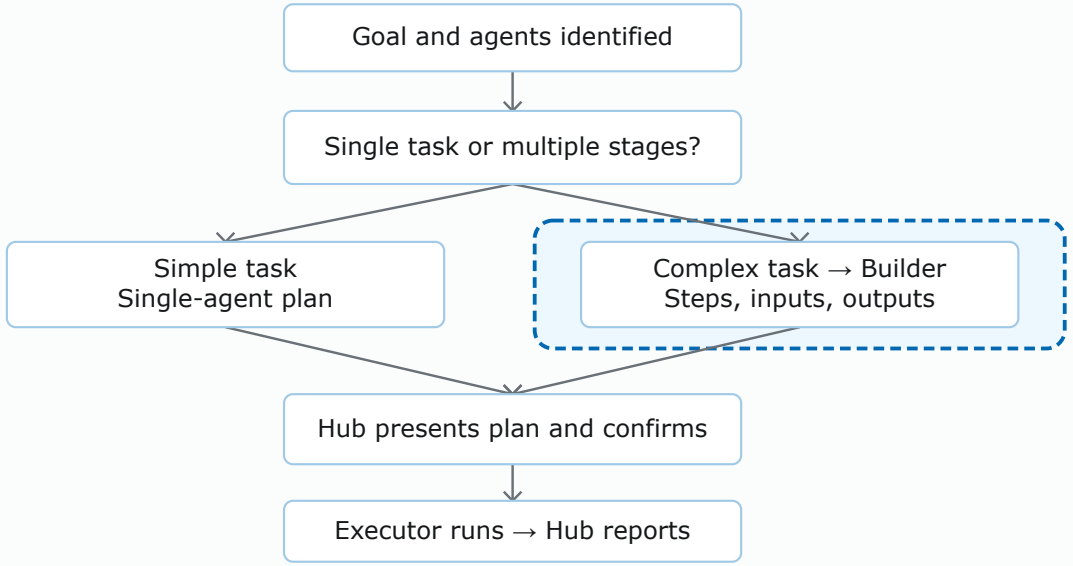
User perspective 3: “My task needs several agents.”

User situation → Ask AI → Complete the rules

01 **User needs several agents**

02 **Ask AI: who plans and who executes?**

03 **Define steps and handoffs**



USER-LED FOLLOW-UP · ILLUSTRATIVE DIALOGUE

USER · Design question (U)

How should several agents work together on one complex task?

AI · Proposed rule (AI)

Break the goal into ordered steps. Specify each step’s owner, inputs, outputs, and dependencies.

USER · Follow-up (U)

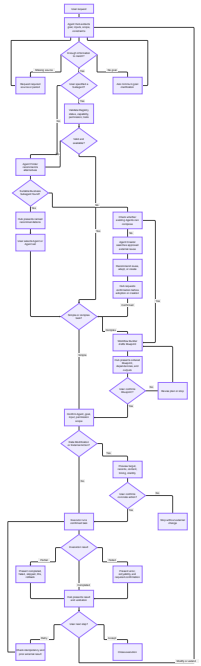
Who plans, who executes, and what exactly passes between them?

IMPLEMENTED IN THE DESIGN

Builder creates the blueprint. Hub presents it for confirmation. Executor receives the confirmed plan and dispatches work within scope.

03 · COMPLETE WORKFLOW

Ask AI to connect the main path, branches, and return paths



Click to explore the full diagram

DESIGNING THE PATH · ILLUSTRATIVE DIALOGUE

USER · Connect the pieces U

Combine our main path and user-question branches into one end-to-end flow. Show where each branch rejoins, waits, or stops.

AI

AI · Trace the return paths

I'll connect the normal path, clarification loops, plan revisions, and result review. I'll identify missing connections rather than invent a decision.

USER · Challenge the map U

If inputs are missing, the plan is rejected, or execution fails, where do we go next? Check each branch for a clear return or stopping point.

Building a flowchart like this helps us check with AI that every branch has a clear next step, return path, or stopping point.

Actual project diagram referenced by README → End-to-End Agent Hub Flow. Overview is for orientation. Open the zoomable view for readable labels. Copy canvas expanded to include source nodes beyond its original bounds.

03 · WORKFLOW

Use a swimlane diagram to clarify who does what

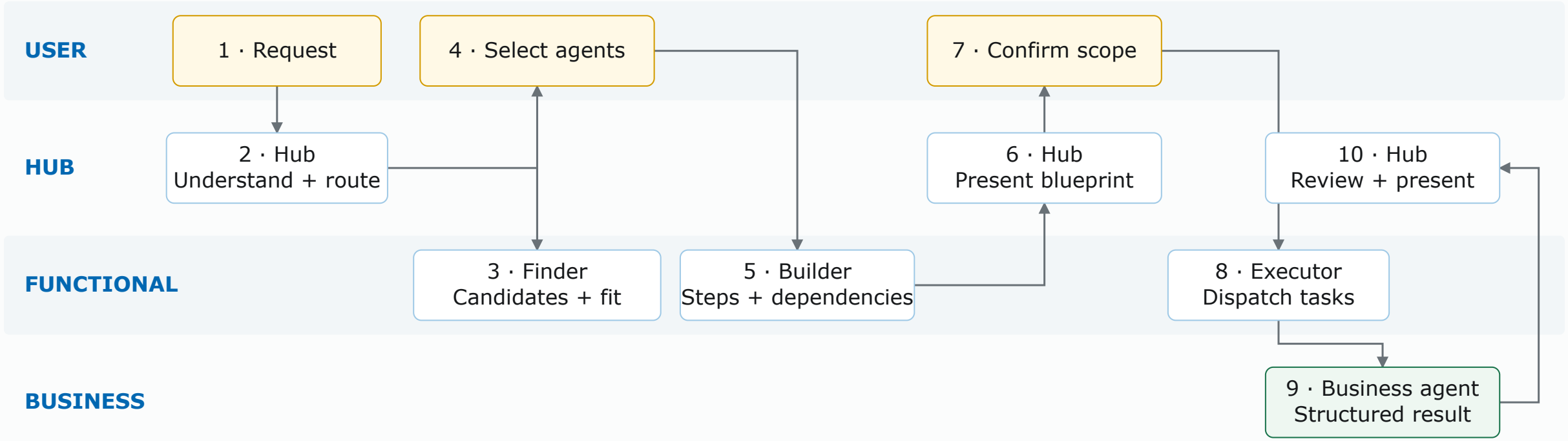
USER — AI · Make the responsibilities explicit

Put each step in the responsible role's lane and define:

Owner: Who is responsible for this step?

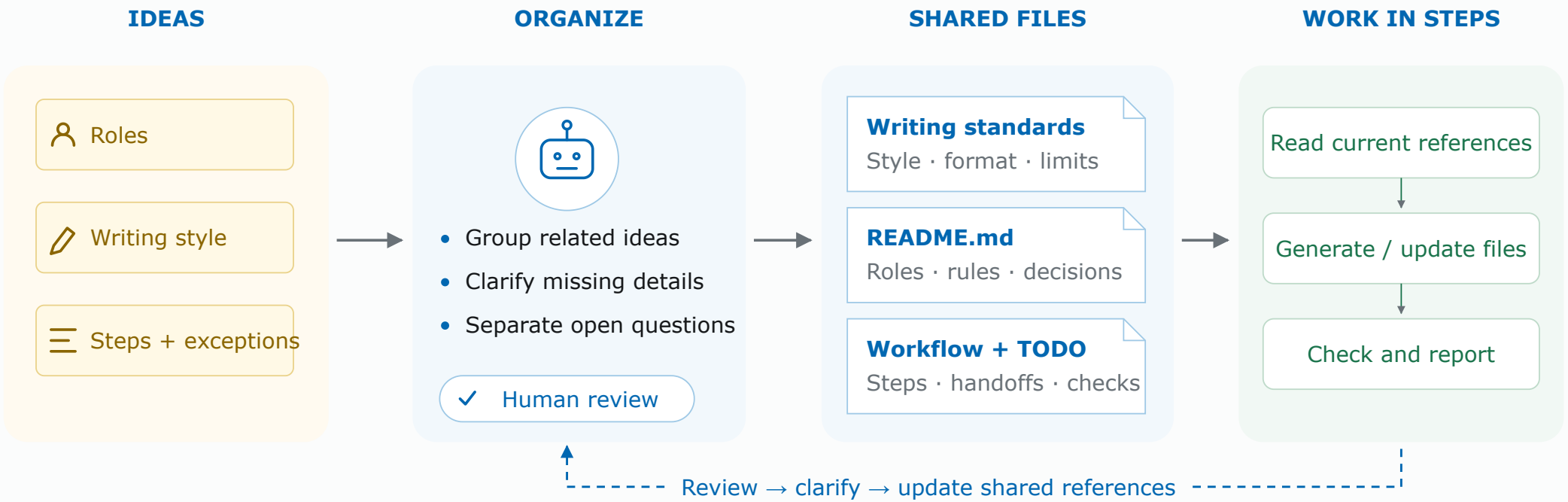
Handoff: What does it receive, produce, and pass to the next role?

Boundary: When should it pause or return? Keep planning separate from execution.



04 · IMPLEMENT WITH AI

Use AI twice: organize our ideas, then work from the agreed files



First: Ask AI to organize scattered ideas into clear roles, workflows, and shared standards.

Second: Ask AI to generate and refine content step by step using the agreed files.

Illustrative collaboration method. We review the organized ideas before treating them as shared rules. File generation is distinct from authorized agent runtime.

04 · IMPLEMENT WITH AI

Four reusable files, four clear jobs

We save the agreed design in separate files, each with a clear purpose: **system design**, **definition standards**, **capability discovery**, or **next steps**. Tell AI which reference to read and which file to update so it can build on our agreement across tasks.



README.md

Shared Design

Roles · Rules · Decisions



Business README

Definition Pattern

Name · Description · Instructions



Registry

Find Capabilities

Tags · Fit · Availability



TODO.md

Track Next Steps

Tasks · Checks · Open Items

Project conventions summarized. Business README: business-subagents/README.md. Registry: Agent-Registry.txt.

04 · IMPLEMENT WITH AI

Ask AI to summarize our agreement, then inspect the README

01 · Ask AI to summarize

USER · Summarize and save

Summarize our agreed roles, workflow, and rules in README.md. Preserve our decisions and flag unresolved questions.

AI · Organize the agreement

I'll group the content into the shared sections, record accepted choices in Decision Log, and keep unresolved items in Open Questions.

02 · Use a fixed framework

Responsibilities

Core Architecture
- ↳ Agent Hub
- ↳ Finder / Builder
- ↳ Executor / Creator

Workflow Rules

Task Routing Logic
- ↳ Browse / Recommend
- ↳ Simple / Complex

Decision Record

Decision Log
- ↳ Accepted Choices

Open Questions
- ↳ Unresolved Choices

03 · Inspect the actual file

agent hub / README.md

Architecture | Finder | Standard | Decisions

Agent Finder

Agent Finder discovers and recommends suitable agents from the enterprise-certified and recommended Subagent resource library.

Use Agent Finder when:

- The user wants to know what agents exist.
- The user searches for agents by category, keyword, goal, or business scenario.
- The user wants the best agent for a task.
- The user wants to execute a task but has not specified the target agent.

Illustrative dialogue. Outline abbreviated for display. Scrollable document is a local snapshot of the actual project README.

Use the business-agent README as a generation standard

business-subagents/README.md

Definition Format

Name + Description
- ↳ English / Chinese Names
- ↳ Description ≤ 800 Characters

Instruction
- ↳ Roles / Triggers / Inputs
- ↳ Outputs / Rules / Limits

Shared Boundaries

Structured Results
- ↳ Return Upstream Only
- ↳ Missing Inputs: needs_clarification
- ↳ assumptions / missing_fields

Consistency Checks

Registry + TXT
- ↳ Keep Metadata Aligned

EXAMPLE · Presentation-Rehearsal-Coach-Agent.txt

Presentation Rehearsal Coach Agent

英文名称：Presentation Rehearsal Coach Agent

中文名称：演讲排练反馈助手

Description

Reviews supplied presentation attempts for clarity, structure, audience fit, and estimated timing, with explicitly simulated audience questions. Use when improving clarity, structure, audience fit, and estimated timing through simulated questions. Returns the requested communication draft or one practice turn with clear assumptions and review notes.

Instruction

Rules summarized from business-subagents/README.md. Right: the actual project definition, available to scroll.

05 · BUILD AND ITERATE

Build one agreed step at a time, then update TODO

USER · Control the implementation pace

Read the current README and TODO. Propose the implementation sequence first. This round, only refine the Hub and Finder definitions. Report checks and pending decisions before continuing.

01



Read the references

README + TODO

Roles · format · open tasks

→

02



Set this round's scope

Hub + Finder only

Name the files and outputs

→

03



Generate and review

Draft → check → revise

Compare with agreed rules

→

04



Save the next starting point

Update TODO

Done · open · next action

After each round, report the files changed, checks performed, and open questions. If the next step requires changing agreed responsibilities or rules, or expanding scope, raise the issue before proceeding.

Illustrative build request based on README's implementation plan and TODO. This concerns generating project files, not authorizing Agent Hub business-task execution.

05 · BUILD AND ITERATE

Resume without losing context or redesigning agreed work

Why restore context?

Lost Context
Earlier decisions are missing from the next request.

Repeated Design
Agreed roles are proposed all over again.

Change Spreads
A small revision rewrites unrelated branches.

01 Read References — 02 Restore Agreement — 03 Locate the Change — 04 Continue Within Scope

READ FIRST README | Workflow | Agent TXT | TODO

USER · Restore context (U)

Read the current README, relevant workflow, Agent TXT, and TODO. Summarize the agreed roles, key rules, completed work, and open items before continuing.

AI · Locate the change (AI)

I'll locate the branch affected by this request, identify the files involved, and explain the proposed change. If the references conflict or a decision is missing, I'll flag it first.

USER · Limit the revision (U)

Update only the agreed scope and preserve unrelated branches. Report the files changed, checks performed, and remaining questions. Ask before changing an accepted role or rule.

Illustrative follow-up. References must be explicitly supplied or accessible. They support consistency checks and do not guarantee retained memory.

05 · BUILD AND ITERATE

Check a revision by walking one concrete request

01 **Choose a request**

CHALLENGE REQUEST

“Use this agent. I’ll send the files later.”

The agent is selected, but required inputs are not ready.

02 **Trace the branch**

Validate the agent
Suitable for this goal?

↓

Check required inputs
Missing materials → pause

↓

Return the missing fields
Hub obtains the needed input

03 **Compare the files**

Compare the written rules

README + Hub
Does routing preserve the input check?

Relevant agent TXT
Are missing fields and pause conditions explicit?

04 **Fix the mismatch**

Targeted feedback to AI

If the files disagree, identify the conflicting clauses and propose a focused correction.

Revise the affected files, then walk through the same request again.

Illustrative design walkthrough, not an executed runtime test. Also check unavailable agents and changes in scope.

05 · ITERATE & VALIDATE

A new idea: fewer repeated prompts, clearer confirmation rules

Two situations need different handling

01 The user has already requested the task

"Summarize the text below."

What we want

With the input ready and scope clear, avoid asking "Should I start?" again.

02 The user adds an external action

"Send the summary to my team."

What we want

Clarify recipients and content, then confirm the specific sending action.

Design goal: remove redundant prompts while keeping necessary clarification and confirmation.

ILLUSTRATIVE CHANGE DISCUSSION

USER · Explain the two situations (U)

If users already ask for a summary and provide the text, I don't want another "Should I start?" But sending it to others is a different action.

AI · Translate the idea into a rule (AI)

I'll distinguish an already requested task from a new external action, then identify where the current confirmation flow would need revision.

USER · Request a reviewable proposal (U)

Show the proposed branches and affected files first. Keep clarification for missing inputs and confirmation for the sending action. Don't change files yet.

Proposed behavior, not implemented. Existing project confirmation rules remain in effect until a revision is accepted.

05 · ITERATE & VALIDATE

If accepted, carry the change through the project files

USER — AI · Apply the accepted change consistently

Record the decision, locate every affected rule and file, update them together, and report checks and open questions.

01 **Record the decision**

Decision Log

Accepted boundary, reason, and unresolved items.

02 **Update shared rules**

README

Confirmation conditions and shared routing rules.

03 **Revise the workflow**

Diagram
Hub / Executor TXT

Affected branches, returns, and execution checks.

04 **Check and save progress**

Checks
TODO

Direct task, missing input, external action, and scope change.

Change the agreed boundary consistently. Preserve unrelated responsibilities and permissions.

Proposed update process. Agent Hub source files and current confirmation policy have not been modified.

05 · ITERATE & VALIDATE

After updating the files, verify what is actually complete

Updating all the files does not mean the feature is ready. Distinguish a completed definition from verified execution and results.

01 · Definition

- Roles and rules are consistent
- TXT and Registry agree

Written definition checked

→

02 · Runtime

- Tools and permissions configured
- Actual agent calls observed

Run evidence needed

→

03 · Results

- Outputs supported by evidence
- Acceptance criteria reviewed

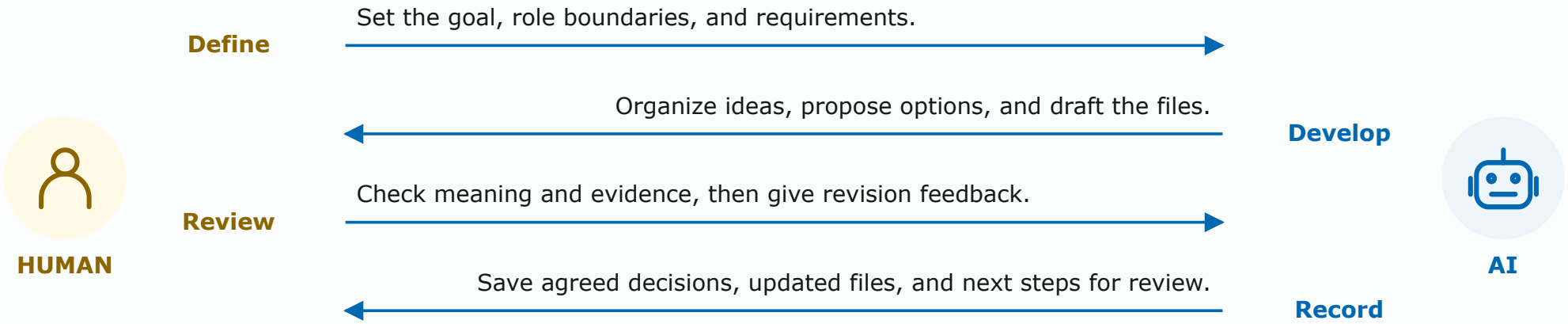
Output review needed

Actual project correction · TODO, 2026-10-15

`Active` means the TXT definition is complete and discoverable. It does not establish deployment or runtime validation.

Slide 17 checks written-rule consistency. This page distinguishes definition, runtime, and result evidence. No claim that all gates have passed.

Human judgment and AI assistance at each iteration



Human judgment guides each iteration.

AI helps organize, develop, and record the work.

A reusable communication template: design the workflow and README

01 **Goal**
What should this achieve?

02 **Context**
Which ideas and files should AI use?

03 **Scope**
What should AI produce this round?

04 **Constraints**
Which rules and boundaries must remain?

05 **Acceptance**
What outputs and checks define completion?

EXAMPLE · ONE COMPLETE REQUEST TO AI

Goal. Help me design Agent Hub, a single entry point for finding and coordinating business agents.

Context. Use my requirements and available project files. First summarize accepted decisions and open questions.

Scope. Map likely user requests into a workflow. Define Hub, functional, and business-agent responsibilities. Propose a minimal file structure and draft the README: vision, architecture, routing, definition standards, decisions, and open questions.

Constraints. Keep recommendation, planning, and execution separate. Require confirmation before execution. Flag missing decisions. Do not invent tools, deployment, or test results.

Acceptance. Return the workflow diagram, file responsibilities, and README draft. Walk through representative user questions and list conflicts, assumptions, and next steps.

> For an existing project: read the current README and diagram first, explain the proposed changes, then synchronize the affected files.

Reusable example derived from Agent Hub’s design. Adapt the five elements to the task. This is not a verbatim historical prompt.

TAKEAWAY

From a conversation to a shared, reviewable design

01

Design the logic

Define roles, decisions, and handoffs before expanding the system.

02

Keep a shared reference

Capture accepted rules in README, diagrams, and agent instructions.

03

Review every change

Check affected files and separate written definitions from verified execution.

Capture the idea as context.

Capture requirements as standards.

Use files to guide AI through implementation and review.